UX Design Brief

Three Portal System — Complete Wireframe Specification

Customer Portal • Business Admin Portal • Platform Admin Portal

65+ Screens • Full Information Architecture

Generated: March 2026

Source: Platform Specification (Epics 01–10)

Comprehensive UX Design Brief — Three Portal System

> **Purpose:** Provide a Figma-ready specification with enough detail for a UX designer to generate wireframes, user flows, and high-fidelity prototypes for every planned screen across all three portals.

> **Source of truth:** Epics 01–10 task plans, architecture docs, workflow specs, API spec, Prisma schema, and tenant lifecycle state machine.

Table of Contents

1. System Overview
2. Design Tokens & Shared Vocabulary
3. Portal 1 — Customer Portal (web-customer)
4. Portal 2 — Business Admin Portal (web-admin)
5. Portal 3 — Platform Admin Portal (web-platform-admin)
6. Cross-Portal Patterns
7. Tenant Lifecycle & Operational Workflows
8. Security & Audit Requirements
9. Deployment & Publishing Workflow
10. Technical Infrastructure Notes for UX
11. Acceptance Criteria Summary

System Overview

Product Scope

A **multi-tenant, template-driven commerce and service operating system** for local businesses (restaurants, retail, services) with three primary user-facing portals operating against a modular monolith backend.

Five Primary Product Surfaces

#	Surface	App Key	Default Port
1	Customer Portal	`web-customer`	4173
2	Business Admin Portal	`web-admin`	4174
3	Platform Admin Portal	`web-platform-admin`	4175
4	Customer App Shell	(PWA infra shared across tenants)	—
5	AI Onboarding & Import Engine	(background worker)	—

Supported Business Verticals

- **Restaurant** — ordering + optional booking
- **Retail** — ordering
- **Service / Salon / Spa** — booking + optional ordering
- **Fitness** — booking (classes)
- **Hybrid / Custom** — any combination of modules

Module System

Tenant features are toggled by modules. Navigation, capabilities, and UI sections dynamically show/hide based on enabled modules:

Module	Controls
Ordering	Cart, checkout, fulfillment, order management
Booking	Availability, slot selection, booking management
Loyalty	Points, tiers, rewards
Content	Pages, SEO, announcements
Analytics	Dashboards, reporting

Design Tokens & Shared Vocabulary

The platform uses a centralized design token system (`@platform/ui`) that provides:

Token Category	Examples
Colors	`--color-primary`, `--color-surface`, `--color-error`, `--color-success`
Typography	`--font-heading`, `--font-body`, `--font-mono`
Spacing	4 px grid: `--space-1` (4 px) through `--space-8` (32 px)
Radii	`--radius-sm` (4 px), `--radius-md` (8 px), `--radius-lg` (16 px)
Shadows	`--shadow-sm`, `--shadow-md`, `--shadow-lg`

Shell States (all portals)

Every portal renders one of five mutually exclusive shell states. Designers should create a screen for each:

State	When	Visual
`loading`	Bootstrap in progress	Centered spinner / skeleton
`error`	Bootstrap failure	Error message + retry CTA
`anonymous`	No auth token	Login / register form
`authenticated`	Logged in, ready	Full app chrome + router outlet
`maintenance`	Platform-wide downtime	Maintenance banner

Component Primitives (shared)

- **AppShell** — Top bar + sidebar nav + content area
- **DataTable** — Sortable, filterable, paginated table with bulk-select
- **FormSection** — Grouped form fields with validation states
- **StatusBadge** — Colored pill (`draft` , `active` , `suspended` , `archived` , etc.)
- **ConfirmDialog** — Modal confirmation for destructive actions
- **Toast** — Success / error / info auto-dismiss
- **EmptyState** — Illustration + CTA for first-time screens
- **SkeletonLoader** — Shimmer placeholder during data fetch

Portal 1 — Customer Portal (web-customer)

Purpose

Tenant-scoped, customer-facing experience for browsing products/services, making purchases/reservations, and managing accounts. Each tenant's storefront loads via **custom domain** or **managed subdomain** with correct branding and configuration.

Information Architecture

```
/ (Storefront Home)
├── /menu (or /catalog)
|   └── /menu/:categorySlug
|       └── /menu/:categorySlug/:itemSlug      → Item detail
├── /services
|   └── /services/:serviceSlug                  → Service detail + book CTA
├── /cart                                        → Cart summary
|   └── /checkout                               → Checkout flow
|       └── /checkout/confirmation/:orderId     → Order confirmation
├── /book
|   └── /book/:serviceSlug                      → Availability picker
|       └── /book/confirmation/:bookingId       → Booking confirmation
├── /account
|   ├── /account/profile                         → Edit profile
|   ├── /account/orders                          → Order history list
|   |   └── /account/orders/:orderId             → Order detail
|   ├── /account/bookings                        → Booking history list
|   |   └── /account/bookings/:bookingId         → Booking detail
|   ├── /account/loyalty                         → Points & rewards
|   ├── /account/payment-methods                 → Saved cards
|   └── /account/notifications                   → Notification preferences
├── /pages/:slug                                 → Content page (About, FAQ, etc.)
├── /login
├── /register
└── /reset-password
```

Screen Inventory & Wireframe Specs

CP-01: Storefront Home

Attribute	Detail
Route	`/`
Purpose	Landing page for tenant storefront
Dynamic elements	Tenant logo, hero image, brand colors, typography — all pulled from tenant config
Sections	Hero banner, featured categories/services, announcements/promotions (if any), business info (hours, address), navigation to catalog/services/booking
Module gating	If ordering disabled → hide catalog CTA. If booking disabled → hide services/book CTA.
Responsive	Mobile-first. Hero stacks vertically. Categories become horizontal scroll.

CP-02: Catalog / Menu Browse

Attribute	Detail
Route	`/menu`, `/menu/:categorySlug`
Purpose	Browse products organized by categories
Layout	Category sidebar (desktop) / horizontal tabs (mobile) + item grid/list
Item card	Image, name, short description, base price, "Add to Cart" button
Search	Search bar filtering by item name
Filters	Category, availability (in stock), price range
Empty state	"No items found" with suggestion to browse all

CP-03: Item Detail

Attribute	Detail
Route	`/menu/:categorySlug/:itemSlug`
Purpose	Full item view with modifiers and add-to-cart
Sections	Image gallery, name, full description, base price, modifier groups (size, add-ons, etc.), quantity selector, "Add to Cart" CTA
Modifier UX	Each modifier group rendered as radio (choice), checkbox (multi-select), or stepper (quantity) based on group type. Required modifiers show validation. Price adjustments shown inline.
Price display	Running total updates as modifiers change

CP-04: Cart

Attribute	Detail
Route	`/cart`
Purpose	Review cart before checkout
Line items	Item name, selected modifiers, quantity, line total, remove/edit buttons
Running total	Subtotal (server-side calculations — no client-side arithmetic trusted)
Actions	"Continue Shopping", "Proceed to Checkout"
Empty state	"Your cart is empty" + CTA to browse menu

CP-05: Checkout

Attribute	Detail
Route	`/checkout`
Purpose	Complete purchase
Sections (sequential)	
1. Fulfillment type	Radio: Pickup / Delivery / In-store (based on tenant & location capabilities)
2. Fulfillment details	Pickup: select location + time. Delivery: enter address + delivery time.
3. Pricing breakdown	Subtotal, modifiers, discounts, tax(location-based), service charges, fees, tip selector (preset %, custom), total
4. Payment	Saved payment method selector **or** new card entry (Stripe/Square embeds). Display tokens only — never show full card numbers.
5. Confirm	Order summary + "Place Order" CTA
Validation	All required fields validated in real-time. Submission blocked until valid. Server revalidation on submit.

CP-06: Order Confirmation

Attribute	Detail
Route	`/checkout/confirmation/:orderId`
Purpose	Post-purchase confirmation
Content	Order number, timestamp, itemized receipt (items + modifiers + pricing breakdown), fulfillment details (location, estimated ready/delivery time), payment confirmation, "View Order Status" link

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CP-07: Services Browse

Attribute	Detail
Route	`/services`
Purpose	Browse available services
Layout	Card grid of services
Card content	Service name, description snippet, duration, base price, "Book Now" CTA
Filters	Category/type, staff, availability

CP-08: Service Detail

Attribute	Detail
Route	`/services/:serviceSlug`
Purpose	Full service info + booking entry point
Sections	Images, full description, duration, pricing (base + add-ons), available staff, cancellation policy, "Book This Service" CTA

CP-09: Booking Availability Picker

Attribute	Detail
Route	`/book/:serviceSlug`
Purpose	Select date, time, staff, location
Step 1	Date picker (calendar widget)
Step 2	Available time slots (computed from location hours, service duration, staff availability, blackout windows, lead time). Unavailable slots shown greyed with reason tooltip.
Step 3	Staff preference (optional or required depending on service config)
Step 4	Location selector (if multi-location)
Step 5	Add-ons selection
Confirmation panel	Service name, date/time, staff, location, pricing, deposit/prepayment terms, cancellation policy
CTA	"Confirm Booking"

CP-10: Booking Confirmation

Attribute	Detail
Route	`/book/confirmation/:bookingId`
Content	Booking ID, service details, date/time, assigned staff, location, cancellation policy + deadline, pricing/deposit info, "View Booking" link

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CP-11: Account Dashboard

Attribute	Detail
Route	`/account`
Purpose	Account overview
Sections	Profile summary, quick links (orders, bookings, loyalty, payment methods), recent activity

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CP-12: Profile Settings

Attribute	Detail
Route	`/account/profile`
Fields	Name, email, phone, address(es), communication preferences (email/SMS/push opt-in per category)
Actions	Save, change password link

CP-13: Order History

Attribute	Detail
Route	`/account/orders`
Layout	Paginated list with: order ID, date, items summary, total, status badge
Filters	Status filter, date range
Actions	View detail, re-order

CP-14: Order Detail

Attribute	Detail
Route	`/account/orders/:orderId`
Content	Full receipt, item breakdown with modifiers, pricing detail (subtotal, tax, fees, tip), fulfillment status timeline (visual stepper), contact/support link

---

CP-15: Booking History

Attribute	Detail
Route	`/account/bookings`
Layout	List with: booking ID, service name, date/time, staff, status badge
Sections	Upcoming (top), Past (below)
Actions	View detail, cancel (if within policy window)

CP-16: Booking Detail

Attribute	Detail
Route	`/account/bookings/:bookingId`
Content	Service info, date/time, duration, staff, location, pricing, cancellation policy + deadline, check-in status
Actions	Cancel (if within window), reschedule

CP-17: Loyalty & Rewards

Attribute	Detail
Route	`/account/loyalty`
Sections	Points balance (large display), tier status + progress bar, points history (earning/redemption log), available rewards catalog with redemption CTAs, tier benefits list
Module gating	Only visible if loyalty module enabled for tenant

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CP-18: Payment Methods

Attribute	Detail
Route	`/account/payment-methods`
Content	List of saved payment tokens (last 4 digits, brand icon, expiry), default indicator, add/remove actions
Security	Display tokens only. Never show full card numbers.

CP-19: Content Page

Attribute	Detail
Route	`/pages/:slug`
Content	Rich text rendered from CMS (About, FAQ, Policies, etc.)
SEO	Meta title, description, OG image applied per page

CP-20: Login / Register / Password Reset

Attribute	Detail
Routes	`/login`, `/register`, `/reset-password`
Login	Email + password, MFA prompt (if enabled), "Forgot password" link, "Register" link
Register	Name, email, phone, password (with strength indicator), terms acceptance
Reset	Email input → confirmation → new password form

CP-PWA: Progressive Web App

Feature	Detail
Installation prompt	Show on supported devices after engagement threshold
Offline support	Cache storefront, catalog (read-only), past orders/bookings
Home screen icon	Tenant-branded app icon
Push notifications	Order status, booking reminders, promotions (opt-in)
Service worker	Static assets, tenant content, user data (last-known-good)

Portal 2 — Business Admin Portal (web-admin)

Purpose

Tenant-scoped operational console for business owners and staff to manage inventory, process transactions, and operate the business.

User Roles & Capabilities

Role	Capabilities
Owner	All capabilities
Admin	`catalog:write`, `orders:manage`, `staff:manage`, `content:publish`
Manager	`catalog:write`, `orders:manage`, `content:publish`
Staff	`orders:manage` (limited order processing only)

> **Module-based narrowing:** Disabled modules further restrict role capabilities. If ordering is disabled, even owners cannot access order management.

Information Architecture

```
/ (Dashboard)
├── /dashboard                              → Overview & key metrics
├── /catalog
│   ├── /catalog/categories                 → Category list + CRUD
│   ├── /catalog/items                      → Item list + CRUD
│   │   └── /catalog/items/:id              → Item edit
│   ├── /catalog/modifiers                  → Modifier groups
│   └── /catalog/import                     → Bulk import/export
├── /services
│   ├── /services/list                      → Service list + CRUD
│   │   └── /services/list/:id              → Service edit
│   └── /services/availability              → Availability rules
├── /orders
│   ├── /orders/active                      → Active orders queue
│   ├── /orders/history                     → Past orders
│   └── /orders/:id                         → Order detail + processing
├── /bookings
│   ├── /bookings/calendar                  → Calendar view
│   ├── /bookings/list                      → Bookings list
│   └── /bookings/:id                       → Booking detail
├── /customers
│   ├── /customers/list                     → Customer list + CRM
│   │   └── /customers/:id                  → Customer profile
│   └── /customers/loyalty                  → Loyalty program config
├── /staff
│   ├── /staff/list                         → Staff member list
│   │   └── /staff/:id                      → Staff detail + schedule
│   └── /staff/scheduling                   → Team availability view
├── /content
│   ├── /content/pages                      → Page list + editor
│   │   └── /content/pages/:id              → Page edit
│   └── /content/announcements              → Promos & announcements
├── /locations
│   ├── /locations/list                     → Location list
│   │   └── /locations/:id                  → Location detail + hours
│   └── /locations/fulfillment              → Fulfillment settings
├── /settings
│   ├── /settings/business-profile          → Business identity & brand
│   ├── /settings/payments                  → Payment gateway config
│   ├── /settings/notifications             → Email/SMS config
│   ├── /settings/domain                    → Custom domain
│   ├── /settings/users                     → Tenant user management
│   │   └── /settings/users/invite          → Invite user
│   └── /settings/integrations              → Third-party integrations
├── /activity                                → Audit / activity log
└── /analytics                               → Analytics dashboard
```

Screen Inventory & Wireframe Specs

BA-01: Dashboard Overview

Attribute	Detail
Route	`/dashboard`
Layout	KPI cards (top) + activity widgets (below) + quick-action cards
KPI cards	Today's sales/orders, pending orders count, upcoming bookings, staff availability status
Widgets	Recent orders list (last 5), recent bookings list (last 5), operational alerts (payment failures, pending approvals)
Quick actions	"Add Item", "Create Booking", "View Orders"
Module gating	Hide ordering widgets if ordering disabled. Hide booking widgets if booking disabled.

BA-02: Business Profile & Brand Configuration

Attribute	Detail
Route	`/settings/business-profile`
Tabs/Sections	Identity, Brand, Media Library
Identity	Business name, legal name, description, contact phone/email, website URL, social media links
Brand	Logo upload + preview, hero image upload, primary & secondary color pickers, typography preset selector, layout template choice
Media Library	Grid of uploaded assets, upload/delete/replace, set featured images
Theme Preview	Live preview panel showing theme changes in real time
Actions	Save, revert to last saved, publish to storefront

BA-03: Location Management

Attribute	Detail
Route	`/locations/list`, `/locations/:id`
List view	Table of locations: name, address, phone, status, fulfillment modes
Detail form sections	
Basic info	Name, address (street/city/state/zip), phone, timezone
Hours	Day-by-day open/close times, closed-day toggle, blackout windows (date range + reason)
Fulfillment	Supported types per location (pickup/delivery/in-store/dine-in). Delivery radius, fee model (flat/percentage/zone), estimated times.
Operating rules	Tax config, service charges, tipping defaults, cancellation policy, payment policy (cash/prepay), lead time, advance ordering window

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BA-04: Category Management

Attribute	Detail
Route	`/catalog/categories`
Layout	Tree/list view with drag-to-reorder
Category fields	Name (unique within tenant), description, display order, featured image, active/inactive toggle, parent category (for hierarchy)
Actions	Create, edit, delete (with child-handling logic), bulk operations

---

BA-05: Item / Product Management

Attribute	Detail
Route	`/catalog/items`, `/catalog/items/:id`
List view	DataTable: name, category, price, status, last edited. Sortable, filterable (category, status, price range, search by name/SKU). Bulk select + bulk edit/delete/export.
Create/Edit form	
Basic info	Name, description (rich text), SKU, product code
Pricing	Base price, cost (for reporting), status (available/sold-out/discontinued), stock hint
Media	Multiple product images, primary image selector, alt text
Categories	Assign to one or more categories, set featured
Modifiers	Add modifier groups (name, type: choice/checkbox/quantity, required flag). Each group has options (name, price adjustment, available toggle). Display order within group.
Actions	Save, activate/deactivate toggle, clone item, preview in storefront
Bulk import	CSV upload with validation report + conflict resolution (duplicate SKUs)

BA-06: Services Management

Attribute	Detail
Route	`/services/list`, `/services/list/:id`
Service form	
Basic info	Name, description, category/type
Duration	Standard duration (minutes), min/max if variable, buffer time between bookings
Pricing	Base price, duration-based pricing (if variable)
Booking settings	Staff assignment (required/optional), deposit (yes/no + amount), prepayment (yes/no), cancellation window (hours), refund policy
Availability	Bookable locations, max daily bookings per location, lead time, blackout windows
Add-ons	Name, price, required/optional
Media	Service images
Status	Active/inactive

BA-07: Staff Management

Attribute	Detail
Route	`/staff/list`, `/staff/:id`
Staff form	Name, email, phone, role/title, bio/specialization, assigned locations, services they can provide, max bookings/day, profile photo, status (active/inactive/on-leave)
Availability calendar	Week/month view. Mark available/unavailable times, working hours per day, time off blocks, recurring patterns.
Booking load	Visual calendar showing booked slots per staff member. Prevent overbooking.

BA-08: Orders Dashboard & List

Attribute	Detail
Route	`/orders/active`, `/orders/history`
Active queue	Grouped by fulfillment type (pickup/delivery). Sorted by promised time. Count badges for pending.
List columns	Order ID, customer name, date/time, fulfillment type, status badge, total, actions
Filters	Date range, status (pending/confirmed/ready/out-for-delivery/delivered/cancelled), fulfillment type, customer search, order ID search
Sorting	By any column

BA-09: Order Detail & Processing

Attribute	Detail
Route	`/orders/:id`
Sections	
Header	Order ID, date, status badge
Customer info	Name, phone, email, delivery address, notes
Items	Line items with quantities, modifiers, unit prices, line totals, special requests
Pricing breakdown	Subtotal, discounts, tax, delivery fee, service charge, tip, total
Fulfillment	Type, requested time, location, delivery address + instructions, estimated times
Payment	Method, status, transaction ID
Status workflow	Buttons progressing: Pending → Confirmed → Ready/Preparing → Out for Delivery → Delivered. Cancel at any stage (with reason). Each transition timestamped.
Refunds	Full or partial refund. Reason selection + notes. Refund method (original payment, credit). Audit trail. Confirmation.
Communication	Send status update (SMS/email). View communication history.

BA-10: Bookings Calendar & List

Attribute	Detail
Route	`/bookings/calendar`, `/bookings/list`
Calendar view	Week/day/month toggle. Color-coded by service or staff. Drag-and-drop cards. Click to open detail. Staff-centric vs. service-centric view toggle.
List view	Booking ID, customer, service, date/time, duration, staff, status badge. Filters: date range, service, staff, status, customer search.

BA-11: Booking Detail & Management

Attribute	Detail
Route	`/bookings/:id`
Sections	
Header	Booking ID, date, status badge
Service	Name, duration, estimated end time
Customer	Name, phone, email, previous booking history
Staff	Assigned staff, reassign option
Location	Where service occurs
Pricing	Service price, add-ons, deposit, total
Notes	Customer special requests, internal admin notes, communication history
Status workflow	Pending → Confirmed → Check-in → Completed. Cancel at any stage (within policy window + reason). Auto-notify customer on status change.
Rescheduling	View available slots, drag-and-drop in calendar, reassign staff. Auto-notify customer.

BA-12: Customers List & CRM

Attribute	Detail
Route	`/customers/list`, `/customers/:id`
List	Name, email, phone, registration date, last activity, loyalty tier. Search by name/phone/email. Filter by date, loyalty tier, segment. Export CSV.
Profile	Basic info, order history (total spent, avg order value, favorites), booking history (frequency, favorite services), loyalty (points, tier, history), communication preferences, admin actions (update info, reset password, adjust points with audit trail, add notes, block/unblock, archive)

BA-13: Loyalty Program Configuration

Attribute	Detail
Route	`/customers/loyalty`
Program config	Points earning rate (per dollar/per transaction), redemption values, tier definitions (name, threshold, benefits, duration), rewards catalog (redemption name, points required, availability, terms)
Management	Batch award points, manual adjustment with reason, promotion creation (bonus points event)

BA-14: Content Pages Editor

Attribute	Detail
Route	`/content/pages`, `/content/pages/:id`
Page list	Title, slug, status (draft/published), last edited, type (about/policy/FAQ/announcement). Filter and sort.
Editor	Title, slug, meta description, rich text editor with media library integration, draft/publish toggle, page type, visibility (public/login-required/staff-only)
SEO fields	Meta title, meta description, slug, OG image, canonical URL
Draft workflow	Save as draft, auto-save, preview, change history
Publishing	Publish live, schedule future publication, unpublish/archive

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BA-15: Announcements & Promotions

Attribute	Detail
Route	`/content/announcements`
Creation	Title, rich text body, image/video, scheduling (start/end date), visibility toggle, target segment (if enabled)

BA-16: Payment Gateway Settings

Attribute	Detail
Route	`/settings/payments`
Connection list	Provider (Stripe/Square), mode (sandbox/production), status (healthy/warning/error), account ID (masked), last verified
Add connection	Select provider → choose environment → secure credential input (API keys or OAuth) → verify (test transaction) → confirm
Connection detail	Status, supported payment methods, recent transaction summary, verify health, disconnect (with confirmation)

BA-17: Tenant User Management

Attribute	Detail
Route	`/settings/users`, `/settings/users/invite`
User list	Name, email, role, status, invited date, last login. Filter by role/status. Search by name/email.
Invite	Email, role selection, permissions preview, send invitation
Edit user	Name, email, role, status (active/suspended), custom module restrictions
Pending invitations	List with resend/cancel actions
Actions	Deactivate (revokes access, keeps records), delete (if no transaction history), view activity trail

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BA-18: Activity & Audit Log

Attribute	Detail
Route	`/activity`
Log entries	Settings changes, user/staff management, catalog/service changes, order/booking changes, content publishing, integration changes
Filters	Action type, actor, entity type, date range
Detail	What changed, who, when, before/after diff
Scope	Tenant-visible events only. No platform-internal or cross-tenant data.

Portal 3 — Platform Admin Portal (web-platform-admin)

Purpose

Platform-owner control plane for creating tenants, managing infrastructure, monitoring health, and operating the multi-tenant platform.

User Roles & Capabilities

Role	Capabilities
Owner	All platform capabilities
Admin	`tenants:read`, `tenants:write`, `domains:manage`, `analytics:read`
Support	`tenants:read`, `impersonation:manage`
Analyst	`tenants:read`, `analytics:read`

> **Key separation:** Platform access ≠ automatic tenant access. Impersonation is explicit, time-bound, and fully audited. Tenant data access only through impersonation or support flows.

Information Architecture

```
/ (Dashboard)
├── /dashboard                              → Platform overview & stats
├── /tenants
│   ├── /tenants/list                       → Tenant list + search
│   │   └── /tenants/:id                    → Tenant detail + lifecycle
│   ├── /tenants/create                     → Provisioning wizard
│   └── /tenants/:id/impersonate            → Impersonation session
├── /domains
│   ├── /domains/list                       → All domains
│   │   └── /domains/:id                    → Domain detail + verification
│   └── /domains/add                        → Register custom domain
├── /publishing
│   ├── /publishing/releases                → Release list
│   │   └── /publishing/releases/:id        → Release detail + rollback
│   └── /publishing/deploy                  → Initiate publish
├── /imports
│   ├── /imports/list                       → Import jobs list
│   │   └── /imports/:id                    → Import detail + review
│   └── /imports/:id/review                 → Review workspace
├── /payments
│   ├── /payments/connections               → Gateway connections
│   │   └── /payments/connections/:id       → Connection detail
│   └── /payments/webhooks                  → Webhook management
├── /notifications
│   ├── /notifications/queue                → Notification delivery queue
│   └── /notifications/webhooks             → Webhook event queue
├── /operations
│   ├── /operations/health                  → System health dashboard
│   ├── /operations/alerts                  → Active alerts + history
│   ├── /operations/jobs                    → Job queue monitoring
│   └── /operations/logs                    → Structured log viewer
├── /audit
│   └── /audit/search                       → Audit log search
├── /analytics
│   ├── /analytics/platform                 → Platform-wide analytics
│   └── /analytics/tenants/:id              → Tenant-specific analytics
└── /config
    ├── /config/templates                   → Vertical template library
    │   └── /config/templates/:id           → Template detail/edit
    └── /config/settings                    → Platform-wide settings
```

Screen Inventory & Wireframe Specs

PA-01: Platform Dashboard

Attribute	Detail
Route	`/dashboard`
KPI cards	Total active tenants, failed payments/webhooks count, pending imports/publishes, system health status (green/yellow/red)
Widgets	Recent tenant activity, active alerts, upcoming renewals, quick links to common actions

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PA-02: Tenant List

Attribute	Detail
Route	`/tenants/list`
Columns	Tenant name, owner email, status badge (draft/active/suspended/archived), created date, last activity, managed domain, custom domain, modules, release version
Filters	Status, vertical type, date range, custom domain yes/no, payment configured yes/no
Search	By tenant name, owner email

PA-03: Create Tenant (Provisioning Wizard)

Attribute	Detail
Route	`/tenants/create`
Step 1 — Business Info	Business name, owner email, vertical/type selection (restaurant, retail, service, salon, fitness, custom)
Step 2 — Modules	Toggle: Ordering, Booking, Loyalty, Content, Analytics
Step 3 — Configuration	Currency, timezone, region
On submit	Creates tenant in `draft` state, assigns owner, generates managed subdomain (`businessname.platform.local`), initializes defaults (theme, nav, tax), creates empty catalog/services, sends owner invitation email

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PA-04: Tenant Detail & Lifecycle

Attribute	Detail
Route	`/tenants/:id`
Tabs	Summary, Configuration, Domains, Releases, Analytics, Impersonation, Audit
Summary	All basic info, contact, vertical, modules, current state
Lifecycle controls	Status badge + transition buttons: Draft→Activate, Draft→Archive, Active→Suspend, Active→Archive, Suspended→Activate, Suspended→Archive. Reason field for suspension/archival. Status change log.
Configuration	Theme info, modules, payment gateway status, custom domain status, last publish date/version

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PA-05: Tenant Impersonation

Attribute	Detail
Route	`/tenants/:id/impersonate`
Start session	Duration selector (1h / 8h / 24h), large confirmation dialog with clear "You are about to impersonate [Tenant Name]" warning
Active session	Launches tenant-admin portal as that tenant. Persistent banner: "IMPERSONATING: [Tenant Name] — Expires in [time]". Easy terminate button.
Tracking	List of active impersonation sessions (who, when, until when). Full audit trail of all actions during impersonation.

PA-06: Domain List

Attribute	Detail
Route	`/domains/list`
Columns	Domain name, tenant, verification status (pending/verified/failed), SSL status (pending/active/error), created date

---

PA-07: Add Custom Domain

Attribute	Detail
Route	`/domains/add`
Form	Domain name input, select associated tenant
Verification instructions	Show required DNS record (CNAME or TXT) with copy-to-clipboard. Step-by-step instructions. Verify button (polls DNS propagation).

PA-08: Domain Detail & Activation

Attribute	Detail
Route	`/domains/:id`
Verification	Status (pending/in-progress/verified/failed) with reason on failure. Retry button.
SSL	Certificate status, expiration date, auto-renewal, issuer details, force renewal button
Pre-activation checks	Verification: ✓/✗, tenant publish health: ✓/✗, tenant status: ✓/✗, payment gateway: ✓/✗. All checks shown with pass/fail.
Promote	Confirmation dialog → activate → monitor → success (primary domain) or fail (rollback + error detail)
Rollback	Revert to prior domain if post-activation health check fails

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PA-09: Releases Dashboard

Attribute	Detail
Route	`/publishing/releases`
List	Tenant name, version, status (pending/active/failed/archived), created date, creator, release notes, validation result

---

PA-10: Publish Workflow

Attribute	Detail
Route	`/publishing/deploy`
Steps	
1. Select tenant	Tenant picker
2. Review changes	Staged catalog/service/content/settings/import changes. Preview.
3. Release notes	Text field
4. Pre-flight checks	Data integrity, config validity, required fields, payment gateway health, domain cert. All pass/fail displayed.
5. Approve	Launch publish. Real-time status stream: validation → building → staging → health check → production.
Result	Success: release live. Failure: error logs + rollback to previous. Notification sent to tenant.

PA-11: Release Detail & Rollback

Attribute	Detail
Route	`/publishing/releases/:id`
Content	Version, status, created date/by, release notes, full changelog from previous, validation report, publishing event log
Rollback	Select prior release → confirmation → rollback → verify restoration. Audit trail recorded.

PA-12: Imports List

Attribute	Detail
Route	`/imports/list`
Columns	Tenant, artifact type (PDF/CSV/image), upload date, status (uploaded/processing/review-required/approved/published/failed), progress indicator
Dashboard widgets	Active count, failed count, recent imports

PA-13: Import Detail & Processing

Attribute	Detail
Route	`/imports/:id`
Pipeline status	Step-by-step: Upload ✓ → Malware scan ✓/✗ → Classification → OCR (progress %) → Extraction → Mapping → Validation
Extracted preview	Source artifact viewer (PDF/image with OCR overlay). Extracted candidates: catalog items, services, staff, content. Confidence scores per field. Source trace (page/section). Errors and warnings.

PA-14: Import Review Workspace

Attribute	Detail
Route	`/imports/:id/review`
Layout	Split-pane: source artifact (left) + staged data (right)
Staged data	Diff view (before/after if updating existing). Low-confidence fields highlighted yellow/orange. Error fields highlighted red. Inline editing.
Quality gates	Block approval if: blocking errors present, confidence below threshold for critical fields, required fields missing. Show specific reasons.
Actions	Approve all, approve partial (select items), reject, mark for revision. Reviewer notes. Confirmation dialog.

PA-15: Payment Connections

Attribute	Detail
Route	`/payments/connections`, `/payments/connections/:id`
List	Tenant, provider, mode, status, last verified, account ID (masked), last transaction
Detail	Provider, mode, account ID (masked), created/verified dates, supported methods, health indicator, recent transaction summary (count, revenue, failures, refunds over 30 days), performance metrics (avg time, volume, failure rate)
Actions	Verify health, re-setup, disconnect

PA-16: Webhook Management

Attribute	Detail
Route	`/payments/webhooks`
Subscriptions	Event types, endpoint URL, last delivery status, delivery success rate
Failures	Event type, timestamp, reason (timeout/auth/5xx), retry attempts, status (pending-retry/permanent-fail). Retry button. Manual resolution.

PA-17: Notification Queue

Attribute	Detail
Route	`/notifications/queue`
Queue	Recipient, type (order confirmation/booking reminder/etc.), status (pending/sent/failed/bounced), scheduled time, retry count
Failures	Bounces, SMS failures, push failures, provider errors. Manual retry. View reason. Mark resolved.

PA-18: Webhook Event Queue

Attribute	Detail
Route	`/notifications/webhooks`
Queue	Event type, source tenant, timestamp, processing status, retry count
Detail	Raw JSON payload view. Retry processing. Mark resolved/ignore.
Failures	Signature verification, handler errors, provider errors. Detailed logs.

PA-19: Health Dashboard

Attribute	Detail
Route	`/operations/health`
System status	API health (green/yellow/red), worker process, database, Redis/cache, external integrations (Stripe, Square, email, SMS) — each with status, last check, response time. Domain DNS/SSL health.
Performance	API response times (p50, p95, p99), throughput (req/sec), error rate, DB query performance, cache hit rate, queue depths (imports, publishes, notifications, webhooks)
Resources	DB connection count, memory utilization, storage by tenant/artifact type, worker load

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PA-20: Alerts & Incidents

Attribute	Detail
Route	`/operations/alerts`
Active alerts	Severity (critical/warning/info), description, affected tenant(s) or global, time started, acknowledge + detail buttons
Alert types	Payment failure threshold, webhook delivery failures, high error rate, performance degradation, resource exhaustion, cert expiration
History	Last 30 days, acknowledge status, duration, resolution actions

---

PA-21: Job Queue Monitoring

Attribute	Detail
Route	`/operations/jobs`
Queue depths	Imports, publishes, notifications, webhooks, exports/backups — count + oldest job age
Job history	Status (pending/running/completed/failed), progress, estimated time, error logs for failures

---

PA-22: Structured Log Viewer

Attribute	Detail
Route	`/operations/logs`
Filters	Severity (error/warn/info/debug), component (API/worker/auth/payments), tenant, time range
Search	Keyword search
Export	Download for external analysis
Request tracing	Trace by correlation ID. Timeline of components hit. Latency breakdown. Errors in trace chain.

PA-23: Audit Log Search

Attribute	Detail
Route	`/audit/search`
Filters	Action (create tenant, publish, impersonate, modify gateway), actor (email), tenant, entity type, status (success/denied/failure), date range, IP address
Results	Timestamp, actor, action, affected entity, result badge, detail button
Detail	Full entry: actor email + ID, action, entity + ID, before/after state, IP/user agent, request ID for tracing, denial reason (if denied)
Sensitive actions	Money movement (refunds, payments, chargebacks), role/permission changes, impersonation sessions, password resets, integration setup/teardown, publish approvals, rollbacks

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PA-24: Platform Analytics

Attribute	Detail
Route	`/analytics/platform`
Tenant metrics	Active count by status, new per period, churn, vertical distribution, module usage %
Transaction metrics	Total orders (count + revenue), bookings (count + value), payment success rate, failed transactions, refund rate, top payment methods
Customer metrics	Total unique, new per period, repeat rate, lifetime value, segmentation
Operational metrics	Import/publish/webhook/notification success rates, avg time to publish, avg time to first order
Performance	API latency (p50/p95/p99), throughput, error rate, DB performance, storage trend

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PA-25: Tenant-Specific Analytics

Attribute	Detail
Route	`/analytics/tenants/:id`
Content	Orders per week/month, revenue trend, customer acquisition, popular products/services, peak hours/days, payment method breakdown, refund rate, booking utilization, retention
Export	CSV/JSON download, scheduled reports

PA-26: Template Library

Attribute	Detail
Route	`/config/templates`, `/config/templates/:id`
List	Template name, description, default modules, themes, created/modified dates
Detail	Name, description, default modules, starter theme, sample catalog/services, sample pages, default settings (tax, tipping). Edit all fields.
Create template	Save current tenant config as new template. Select which modules/config to include.
Apply	Used during tenant provisioning (PA-03)

PA-27: Platform Settings

Attribute	Detail
Route	`/config/settings`
Sections	Default currency/timezone, supported payment providers, notification providers, analytics provider, security (MFA requirements, session timeout), rate limiting thresholds, feature flags

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Cross-Portal Patterns

Authentication Flows

Portal	Method	Details
Platform Admin	Strong auth required	MFA support/enforcement, API tokens with scopes, shorter session timeouts, failed-login rate limiting
Tenant Admin	Standard email/password	MFA optional, auto-logout on inactivity, "remember this device"
Customer	Email or phone registration	Password reset, optional MFA, session tokens for mobile/PWA

Validation & Error Handling

Pattern	Behavior
Field validation	Real-time, clear messages, field-level + cross-field, prevent submission until valid, server revalidation
Error states	Plain language (not technical jargon), actionable next steps, contact support link, error codes for support, retry where appropriate, fallback for non-critical failures

Loading & Progress

Pattern	Usage
Skeleton screens	Initial page loads
Progress indicators	Long operations (imports, publishes)
Real-time status	WebSocket or polling for order/booking/job status
Estimated time	Shown where calculable
Cancel option	Available for long-running operations

Empty States

Every list/table screen needs a designed empty state:

- Helpful message explaining what will appear here
- Contextual CTA ("Create your first item", "Upload first artifact")
- Explanatory graphic/illustration
- Link to documentation

Mobile Responsiveness

Aspect	Approach
Layout	Mobile-first across all portals
Touch	Touch-friendly elements (min 44 px tap targets)
Tables	Collapsible/card view on mobile
Navigation	Sidebar → hamburger drawer
Parity	Full feature parity on mobile where practical

Accessibility (WCAG 2.1 AA)

- Keyboard navigation throughout
- Screen reader support
- Color contrast ratios meeting AA
- Visible focus indicators
- ARIA labels
- Form labels properly associated with inputs

Notifications

Channel	Use Cases
Toast (in-app)	Action feedback (success/error/info), auto-dismiss or manual close
Modal alert	Critical confirmations (delete, cancel, refund)
Email	Order confirmations, booking confirmations, status updates, admin alerts
SMS	Order status, booking reminders, account alerts
Push (PWA)	Order status, booking reminders, promotions (opt-in)

Search & Filtering

- Global search within current scope
- Quick filters (most common visible by default)
- Advanced search with multiple criteria
- Filter persistence (remember last used)
- Saved searches (where applicable)
- Export results

Tenant Lifecycle & Operational Workflows

State Machine

```
┌───────┐      activate      ┌────────┐
│ Draft │──────────────────→│ Active │
│       │                    │        │
└───┬───┘                    └──┬─┬───┘
    │                          │ │
    │ archive     suspend ─────┘ │ archive
    │             │              │
    ▼             ▼              ▼

┌─────────┐  ┌──────────┐  ┌─────────┐
│ Archived │  │ Suspended │  │ Archived │
│(terminal)│  │           │  │(terminal)│
└─────────┘  └──┬─┬──────┘  └─────────┘
                │ │
          activate ┘ └ archive
                     │
                     ▼

                  ┌─────────┐
                  │ Archived │
                  │(terminal)│
                  └─────────┘
```

Operational rules:

- Draft tenants: allowed through setup flows, blocked from live routing
- Suspended/archived tenants: fail closed before data access
- All lifecycle mutations are audit-logged
- Dependent modules don't invent their own transitions

Security & Audit Requirements

Data Protection

- Payment gateway secrets encrypted at rest
- Webhook secrets encrypted at rest
- Never echo back sensitive values in UI
- Customer PII redacted in logs
- Database passwords and API keys never in logs
- Audit trails record sensitive actions without exposing values

Access Control Evaluation

Tenant-Admin path: Actor type = tenant → User ID present → Active membership → Role allows capability → Module doesn't narrow further

Platform-Admin path: Actor type = platform → User ID present → Valid platform role → Role allows capability → Tenant access denied unless explicit impersonation

Security Events to Surface in UI

- Failed authentication attempts
- Rate limiting triggers
- Unauthorized access attempts
- Suspicious activity detection

Deployment & Publishing Workflow

Create Business End-to-End Flow (10 Stages)

Stage	Owner	Key Actions
1. Provisioning	Platform Admin	Create tenant (draft), select template/modules, generate subdomain, send owner invite
2. Brand & Identity	Tenant Owner	Upload logo/assets, configure colors/typography, define legal metadata
3. Operational Config	Tenant Owner	Configure locations/hours, set fulfillment, define tax/pricing, invite staff
4. Payments	Tenant Owner	Connect gateway, verify credentials, configure payouts
5. Data Ingestion	Tenant Owner / Auto	Upload menu/catalog/services (PDF, CSV, image, form)
6. AI Extraction	Automation	OCR, classification, entity extraction, confidence scoring, conflict detection
7. Review & Approval	Reviewer / Owner	Staged data review workspace, corrections, approval gate
8. Publish	Automation / Owner	Generate storefront config, publish to preview, smoke checks, publish to production, elevate to active
9. Domain Activation	Platform Admin / Owner	Verify custom domain, health check, promote domain, verify routing + cert
10. Post-Publish	Automation	Seed analytics, schedule backups, enable monitoring, record completion

Failure Handling Principles

- Import failures → resumable jobs
- Publish failures → preserve last known good config
- Domain activation failures → don't block managed subdomain
- All failures are observable in operator views

Technical Infrastructure Notes for UX

Request Routing Model

Domain	Resolves To
Platform admin domain	Platform admin portal
Verified custom domain	Customer portal (correct tenant)
Managed subdomain	Customer portal (correct tenant)
Preview environment	Access-controlled preview
Admin portals	Always require authentication + tenant membership

Asynchronous Workflows

Designers should plan for async status patterns:

- Import processing → background job → real-time progress + results
- Publishing → async pipeline → status stream
- Notification delivery → queued → delivery status
- Webhook processing → idempotent queue → processing status

Real-Time Features

- WebSocket for status updates (orders, bookings, jobs)
- Polling fallback for unsupported browsers
- Push notifications for critical events
- Real-time analytics dashboards

Acceptance Criteria Summary

Customer Portal

- [ ] Tenants render branded storefronts without code forks
- [ ] Customer ordering and booking flows work end-to-end
- [ ] Account history and loyalties persist correctly per tenant
- [ ] App shell is installable as PWA with offline support
- [ ] Authentication and session management are secure

Business Admin Portal

- [ ] Tenant owners configure business without platform intervention
- [ ] Admins only see their own tenant data

- [ ] Configuration changes propagate to storefront safely
- [ ] Order and booking operations are intuitive and complete
- [ ] Staff and customer management are full-featured

Platform Admin Portal

- [ ] Platform admins can provision tenants and manage infrastructure
- [ ] Impersonation is explicit, auditable, and time-bounded
- [ ] Publishing is versioned, reproducible, and reversible
- [ ] Integration failures (payments, webhooks, domains) are observable
- [ ] Audit trails cover all privileged actions and money movement

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Screen Count Summary

Portal	Screen Count	Key Wireframes
Customer Portal	20 + PWA	CP-01 through CP-20
Business Admin Portal	18	BA-01 through BA-18
Platform Admin Portal	27	PA-01 through PA-27
Total	**65+**	

Generated from platform specification (Epics 01–10, architecture docs, workflow specs, Prisma schema, API spec, and tenant lifecycle state machine).